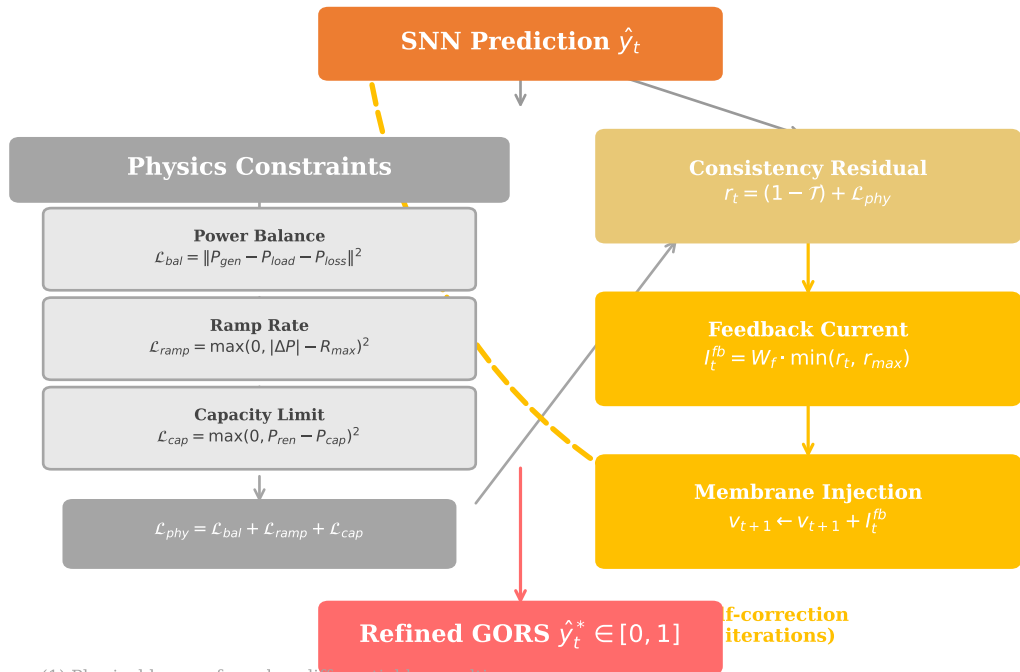


Physics-constrained Learning with Unsupervised Closed-loop Refinement



(1) Physical laws enforced as differentiable penalties

(2) Internal consistency residual drives self-correction

(3) No ground-truth labels required for refinement